Estimation of a score-explained non-randomized treatment effect in fixed and high dimensions

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In non-randomized treatment allocation models, treatment is assigned to a unit based on a score, e.g., scholarship is allocated based on the score obtained in a merit test, while antihypertensive treatments are allocated based on blood pressure level. In this paper, we present a new model coined SCENTS: Score Explained Non-Randomized Treatment Systems that utilizes the dependency of the score on the explanatory variables to permit efficient estimation. We derive an estimator of the treatment effect which is $\sqrt{n}$ consistent, asymptotically normal, and achieves semiparametric efficiency under normal errors. The analysis is extended to ultra-high dimensional vectors of covariates, where a $\sqrt{n}$ consistent and asymptotically normal debiased estimator is proposed. We analyze two real data sets via our method and compare our results with those obtained by using previous approaches like regression discontinuity design. Some possible extensions are discussed.

Keywords: Confounding variables; endogeneity; global effect; high dimensional analysis; semiparametric efficiency

1. Introduction and background

Estimating a treatment effect under non-random treatment allocation has been extensively studied in the statistics, biomedicine, and econometrics literature. To introduce the idea, imagine that a scholarship-granting agency tests a group of high school students and assigns scholarships to those whose scores are above some pre-determined cutoff. It is of interest to determine whether the scholarship has any tangible outcome on future academic performance. Letting Y_i be some measure of the performance of student i in a subsequent semester, we can write down a model of the type:

$$Y_i = \alpha_0 \mathbb{1}_{Q_i \geq \tau_0} + X_i^\top \beta_0 + \nu_i \,, \tag{1.1}$$

where X_i is an (observed) covariate vector including demographic information on the students, Q_i is the score on the scholarship test, τ_0 is a pre-determined known cutoff, and ν_i is a residual term. Observe that one can take without loss of generality $\tau_0 = 0$ by defining $Q \leftarrow Q - \tau_0$. The parameter α_0 represents the effect of the treatment: scholarship. If a statistical test rejects the hypothesis $\alpha_0 = 0$, one concludes that the scholarship significantly impacts subsequent academic scores. A simple regression analysis cannot be performed to estimate the treatment α_0 owing to the possible dependence between Q and ν. Similar examples are also prevalent in biomedicine, especially when assessing the efficacy (or possible side effects) of existing drugs. For example, quantifying the side effects of a statin prescribed when total cholesterol exceeds a certain threshold.

It is instructive to compare and contrast this setup with randomized clinical trials. In both, we estimate the effect of a treatment that applies to a subset of the participants. The main difference is that the treatment allocation in the latter case is independent of the covariates and error terms in the model. In contrast, the applied treatment is non-trivially correlated with the covariate and error terms

in the former case and is, therefore, endogenous. In our example, the score Q_i on the assessment test, based on which the treatment (scholarship when $Q_i > 0$) is applied, is correlated with Y_i beyond the shift-effect α and the background covariates X_i, as the unmeasured sources of variation (say, native abilities/genotypical characteristics of the individual not captured by the observed covariates) that are contained in the residual, ν_i, can also contribute non-trivially to the score Q_i on the assessment test. (More meritorious students are more likely to get the scholarship and are also more likely to have higher values of Y over and above what is predicted by the scholarship effect and their demographics!) This typically translates to what is called the 'endogeneity assumption': $E(\nu_i|Q_i) \neq 0$, and hence the model in equation (1.1) cannot be estimated using simple linear regression of Y on $(\mathbb{1}_{Q\geq 0}, X)$, unlike the randomized trial framework. Another model that addresses the endogeneity problem was initially studied under the name of regression discontinuity design (RDD) in (Thistlethwaite and Campbell (1960), Campbell and Stanley (1963)), where data from a national merit competition was analyzed (see Holland and Stalnaker (1957) and Thistlethwaite (1959)). The notion of RDD has since found varied applications, for example, Stadthaus (1972), Erickson and Cromack (1972) in education, Lohr (1972) in health-related research, Berk and Rauma (1983) in social research, Finkelstein, Levin and Robbins (1996) in epidemiology, to name just a few. The main solution to the problem was to compare students whose scores fall in the vicinity of the threshold: specifically, comparing students with $Q_i \in (-h, 0)$ to those with $Q_i \in (0, h)$, where h is some small number that asymptotically converges to 0. The heuristic assumption is that restricting attention to students whose scores are apriori almost equal but on different sides of the threshold corresponds to an almost random assignment and can, therefore, be taken to approximate an RCT tolerably well.

The main drawback of this *local analysis* is it often selects a (data-driven) bandwidth h_n and considers the observations $Y_i's$ for which $|Q_i - \tau_0| \leq h_n$. Only these selected Y_is are then used to estimate the magnitude of the jump discontinuity. As this approach effectively uses $O_p(nh_n)$ observations for estimation purposes, the rate of convergence of the estimator is *slower that* $\sqrt{n}$ (typically $\sqrt{nh_n}$, where the bandwidth $h_n \to 0$ is chosen by standard bias-variance trade-off), see, for example Calonico et al. (2019). Although the local analysis is quite general, i.e. it can incorporate heterogeneity in the treatment effect (α may depend on (X, Q)), the precision with which the homogeneous treatment-effect in equation (1.1) (or a structured heterogeneous treatment effect elaborated in Section 6) is estimated can be improved in situations where it is possible to exploit the relationship between the score Q and the measured covariates X on the individuals of interest. The aim of this manuscript is to propose and analyze a new model that captures this overall phenomenon by suitably using information from all the samples. In such cases, as we argue below, it is possible to estimate the treatment effect at $\sqrt{n}$ rate.

Our contribution: In many practical problems, the score variable Q can be (partially) explained by some observed covariates. For example, in the case of scholarship allocation based on performance in a merit test, the score of the participants relies on their socio-economic background that influences their upbringing. This motivates us to augment (1.1) by a second equation that explains the score Q through the background covariates Z:

$$\begin{aligned} Y_i &= \alpha_0 \mathbb{1}_{Q_i \geq 0} + X_i^\top \beta_0 + \nu_i \,, \\ Q_i &= Z_i^\top \gamma_0 + \eta_i \,. \end{aligned} \tag{1.2}$$

This is the main estimating equation of this paper. The effect of unmeasured covariates that can affect both Q and Y is encoded by the mean 0 error vector (ν_i, η_i).

Assumption 1.1. *The errors (η, ν) are independent of (X, Z) and have zero expectation.*

However, a weaker mean-independent assumption (see Assumption 3.1) is sufficient; see Remark 3.2. We *do not* place any parametric assumptions on the distribution of error (η, ν), i.e., they may be dependent and $b(\eta) \equiv E(\nu|\eta) \neq 0$, where $b(\cdot)$ can be any non-parametric function. Furthermore, X and Z may be identical, completely disjoint, or share several covariates. The intuition behind keeping X and Z different is that some extra covariates may be available while measuring Y but not Q or vice-versa. Various extensions of this model are possible and are discussed in Section 6. In this paper, we concentrate on the model mentioned above.

At first glance, the augmentation of the second equation in (1.2) may suggest that Z is used as an instrumental variable. However, this is not necessarily the case in our model since the *exclusion restriction* Lousdal (2018) that is critical in the instrumental variables approach, namely that the instrument should influence Y only through Q and not directly, is no longer satisfied (especially when $X = Z$, which is typically the case for many real applications). Consequently, the two-stage least squares procedure Angrist and Imbens (1995) typically employed in the instrumental variable literature is *invalid* in our problem. We emphasize that, in contrast to instrumental variable regression, we *do not need* to deploy a completely new set of covariates to explain Q; rather, we can typically use at least part of the already available background information X that influences Y also to explain the score Q. This is typically the case in many real-life examples, e.g., in the scholarship example, the background information that explains the future academic performance of a student (Y) also influences their scholarship test score Q.

We next discuss how this augmentation helps us obtain a $\sqrt{n}$-consistent estimator. *If we observe* η, writing $\nu_i = b(\eta_i) + \epsilon_i$, our model would reduce to a simple partial linear model (see e.g., Yatchew (1997), Bhattacharya and Zhao (1997), Müller and Van de Geer (2015) and references therein) with (α_0, β_0) being the parametric component and the unknown mean function b being the nonparametric component:

$$Y_i = \alpha_0 \mathbb{1}_{Q_i \geq 0} + X_i^\top \beta_0 + b(\eta_i) + \epsilon_i \,. \tag{1.3}$$

Hence, we could acquire a $\sqrt{n}$ consistent estimator of α_0 following the standard analysis of the partial linear model *provided* $\mathbb{E}(\mathsf{var}((\mathbb{1}(Q > 0)\ \mathbf{X}) \mid \eta)) > 0$ (for more details about semiparametric efficient estimation in partial linear models, see Robinson (1988) or Schick (1986)). Indeed, without this assumption, it is easy to see that there is an identifiability problem, and the effect of α_0 cannot be separated from the effect of b in the above model.

As *we do not observe* η, we cannot use the standard partial linear model analysis proposed in Robinson (1988) or Schick (1986) to estimate α_0. We can nevertheless approximate it by the second equation of (1.2), using the residuals obtained via regressing Q on Z, i.e., plug-in the estimates of η_is in equation (1.3) and treat the resulting equation as an *approximate partial linear model*. Indeed, this idea lies at the heart of our method. However, a naive replacement of η by $\hat{\eta}$ (obtained by regressing Q on Z) is not sufficient, as the approximation error of η is $\hat{\eta} - \eta = O_p(n^{-1/2})$, while we need the approximation error to be of the order $o_p(n^{-1/2})$ for $\sqrt{n}$-consistent estimation of α_0. Hence, some more fine-tuning is needed to remove the bias. We elaborate our method of estimation in Section 2.1.

The assumption $\gamma_0 \neq 0$ is critical to our analysis because if not, $Q_i = \eta_i$ and equation (1.3) becomes:

$$Y_i = \alpha_0 \mathbb{1}_{Q_i \geq 0} + X_i^\top \beta_0 + b(Q_i) + \epsilon_i \,.$$

It is now no longer possible to estimate α_0 at $\sqrt{n}$ rate as it is hard to separate the first and third terms of the right-hand side of the above equation. Thus, the augmenting equation $Q = Z^\top \gamma_0 + \eta$ with $\gamma_0 \neq 0$ is what prevents the variable corresponding to the parametric component of interest from becoming a measurable function of the variable corresponding to the non-parametric component of the model, and enables converting the estimation problem to an *approximate* partial linear model. As noted earlier,

for α_0 to be estimated as $\sqrt{n}$ rate, Q does not need to have a linear relation with Z; in fact, any non-linear parametric relation like $Q = g(Z) + \eta$ for some known link function g will work as long as $\mathbb{E}(\mathsf{var}(\mathbb{1}(Q > 0) \mid \eta)) > 0$. Also the assumption $(\nu, \eta) \perp\!\!\!\perp (X, Z)$ is not necessary, all we need is that $\mathbb{E}[\eta \mid Z] = 0$ and $\mathbb{E}[\epsilon \mid X, \eta] = 0$ for $\sqrt{n}$-consistent estimation of α_0.

There is also a degree of similarity between our model and *triangular simultaneous equation models*, which are well-studied in the economics literature. Interested readers may look at Newey, Powell and Vella (1999), or Pinkse (2000) and references therein for more details. However, there is a key difference: the triangular simultaneous equation models generally assume a smooth link function between Y_i and (X_i, Z_i, Q_i), whilst our model presents an inherent discontinuity. Hence, our work *cannot be derived* from analyses of the triangular simultaneous equation model.

Estimation of the treatment effect via partial linear model in the context of regression discontinuity design has also been mentioned in Porter (2003) through the lens of the following model:

$$Y_i = \alpha_0 \mathbb{1}_{Q_i > 0} + b(Q_i) + \epsilon_i, \quad \mathbb{E}[\epsilon \mid Q] = 0\,.$$

The author argued that one *cannot estimate* α_0 at $\sqrt{n}$ rate as $S_i = \mathbb{1}_{Q_i > 0}$ can be completely explained by Q_i, i.e. $\mathbb{E}[\mathsf{var}(S \mid Q)] = \mathbb{E}\left[(S - \mathbb{E}[S \mid Q])^2\right] = 0$ (see the discussion after equation (1.3)). The main difference between our model and that of Porter (2003) is the usage of the background information X (or (X, Z)), which influences both Q and Y. This, on the one hand, prevents us from using X as simple instruments and, on the other hand, ensures that S is not completely explained by η (the random variable corresponding non-parametric component b) as it also depends on X, which enables us to estimate the parametric component at $\sqrt{n}$ rate.

Recently Angrist and Rokkanen (2015) presented an idea where one may exploit background covariates to estimate the treatment effect at $\sqrt{n}$ rate under the assumption $\mathbb{E}[Y \mid Q, X] = \mathbb{E}[Y \mid X]$, i.e., the relation between Q and Y can be explained fully by X, which permits them to perform simple OLS method to obtain a $\sqrt{n}$-consistent estimator of the treatment effect. However, their assumption is not satisfied in our case: the main component of our model is the unobserved η that quantifies the *innate* ability of an individual (i.e., intelligence), which, along with X's, influences both Q and Y (in-fact it might even be argued that η is as important as X, especially in the scholarship example like one presented in Angrist and Rokkanen (2015), as innate abilities play a primary role in the performance of an individual in the scholarship test and beyond).

Extension to high dimensional setup: We extend our analysis to the high dimensional model, assuming that the dimension of both X and Z is much larger than the available sample size. Analysis of treatment effect in the presence of high dimensional covariates is a relatively new topic. For example, Belloni, Chernozhukov and Hansen (2014) proposed a debiased approach for inference on the treatment effect in the presence of high dimensional controls, and, recently, Qiu, Tao and Zhou (2021) studied heterogeneous treatment effect in the presence of high dimensional covariates under the availability of instrumental variables; Guo, Zhou and Ma (2021) proposed a modeling strategy for personalized medicine in the presence of high dimensional covariates and Arai et al. (2019) presented an approach to estimate the parameters of a standard regression discontinuity design in high dimensions.

Following the previous discussion, our model in the high dimensional setup can be viewed as a variant of the high dimensional partial linear model where we do not observe the random variable corresponding to the non-linear part, but instead, a noisy version of it, while allowing the noise to be correlated with the random variables corresponding to the linear and non-linear components of the model simultaneously. Several efforts have been made to estimate both the linear and the non-linear parts of a high-dimensional partial linear model, of which we mention a few here. Lian, Zhao and Lv (2019) used a projected spline estimator along with the ℓ_1 penalty to estimate the parametric and non-parametric parts of a high dimensional partial linear model; Lv and Lian (2017) extended the analysis

to the distributed setup; Müller and Van de Geer (2015) used the ℓ_1 penalty for the linear part and a smoothing penalty (i.e. penalty involving the double derivative of the non-parametric function) for the non-linear part; Han, Ren and Zhu (2017) proposed an adaptive approach based on Honoré and Powell (2005)'s pairwise difference based method which does not require knowledge of any function class apriori; Zhu, Yu and Cheng (2019) proposed a method to debias the estimator of the parameter corresponding to the linear part which facilitates inference for any particular co-ordinate or some linear combination of the parameters corresponding to the linear part. To the best of our knowledge, the closest paper that studies a similar situation (i.e., a partial linear model with erroneous observations corresponding to the non-linear part) is Zhu (2017). Our analysis is different from that of Zhu (2017) for the following reasons: (i) our model is quite different from that considered in Zhu (2017) and hence so is our estimation technique and (ii) we provide a debiased estimate of the treatment effect α_0 and establish asymptotic normality for purposes of inference, whereas Zhu (2017) only calculate the rate of the estimation error in their model. To summarize, our work makes the following contributions:

1. We are able to take care of endogeneity among the errors in a general manner and provide a $\sqrt{n}$-consistent estimator of the treatment effect. Indeed, the main feature of our approach lies in modeling Q itself in terms of covariates up to error terms, which enables *the use of the entirety of data available* and *not just the observations in a small vicinity of the boundary defined by the Q threshold*, on which many existing approaches are typically based.
2. Our estimate achieves semiparametric efficiency under an appropriate submodel.
3. Our method is insensitive to the values of the tuning parameters. As will be seen in Section 2.1, we do require tuning parameter specifications for the non-parametric estimation of $b(\eta)$, but as long as those parameters satisfy some minimal conditions, our estimate of the target parameter α_0—in terms of both rate of convergence and asymptotic distribution—does not depend on them.
4. Our analysis of SCENTS in the presence of high dimensional covariates appears to be the first systematic attempt to deal with debiasing in the high dimensional partial linear model where we observe a noisy version of the random variable corresponding to the non-linear part and the noise is correlated with the random variables involved in the linear part as well as the non-linear parts.

Organization of the paper: In Section 2, we describe the estimation procedure of the treatment effect along with theoretical guarantees when the dimension of the observed covariates is fixed, i.e., does not change with the sample size n. In Section 3, we extend our methodology and analysis when the number of covariates is much larger than the sample size. We present some simulation results to corroborate our theoretical findings in Section 4. In Section 5, we provide analyses of two real data examples using our method and comparisons to existing methods. In Section 6, we present some possible future research directions based on this work. Rigorous proofs of the main results and additional details can be found in the supplementary document Mukherjee, Banerjee and Ritov (2023).

Notation: For any matrix A, we denote by $A_{i,*}$, the ith row of A and by $A_{*,j}$ the jth column of A. Both $a_n \lesssim_P b_n$ and $a_n = O_p(b_n)$ bear the same meaning, i.e. a_n/b_n is a tight (random) sequence. Also, for two non-negative sequences $\{a_n\}$ and $\{b_n\}$, we denote by $a_n \gg b_n$ (respectively $a_n \ll b_n$), the conditions that $\liminf_n a_n/b_n \to \infty$ (respectively $\limsup_n a_n/b_n \to 0$). For any random variable (or object) X, we denote by $\mathcal{F}(X)$ the sigma-field generated by X. For two random variables X and Y, $X \perp\!\!\!\perp Y$ indicates that X and Y are independent, and $X \perp Y$ denotes that they are uncorrelated.

2. Estimation procedure for α_0 in fixed dimension

2.1. Methodology

As mentioned in Section 1, our model for estimating the treatment effect can be written as:

$$Y_i = \alpha_0 \mathbb{1}_{Z_i^\top \gamma_0 + \eta_i \geq 0} + X_i^\top \beta_0 + \nu_i \,,$$

where ν_i and η_i are correlated. Defining $b(\eta_i) = \mathbb{E}(\nu_i \mid \eta_i)$, write $\nu_i = b(\eta_i) + \epsilon_i$ where $\epsilon_i \perp \eta_i$. Using this, we can rewrite our model as follows:

$$Y_i = \alpha_0 \mathbb{1}_{Z_i^\top \gamma_0 + \eta_i \geq 0} + X_i^\top \beta_0 + b(\eta_i) + \epsilon_i \,. \tag{2.1}$$

We first divide the whole data into three (almost) equal parts, say $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$. Henceforth, for simplicity, we assume each $\mathcal{D}_i$ has $n/3$ observations. Denote the dimension of X and Z by p_1 and p_2, respectively. The first two data sets are used to obtain (consistent) estimates of several nuisance parameters, which are then plugged into the equations corresponding to the third data set from which an estimator of the treatment effect is constructed. The data splitting technique makes the theoretical analysis more tractable as one can use independence among the three subsamples to our benefit. Furthermore, by rotating the samples (to be elaborated below), we obtain three asymptotically independent and identically distributed estimates, which are then averaged to produce a final estimate that takes advantage of the full sample size. While we believe that the estimator obtained without data-splitting achieves the same asymptotic variance, a point scorroborated by simulation studies (Section 4), an analysis of this estimator would be incredibly tedious with minimal further insight.

At this point, one may wonder why it is possible to identify α_0 in the presence of endogeneity. Observe that, as ν and η are correlated, the so-called *unconfoundedness* assumption, prevalent in causal inference, is violated. To elaborate, we first align our model with the notations of standard potential outcome framework; suppose $Y(1)$ denotes the response under treatment and $Y(0)$ denotes the response under control. Then, under model (1.2), we have:

$$\begin{aligned} Y(1) &= \alpha_0 + X^\top \beta_0 + b(\eta) + \epsilon \,, \\ Y(0) &= X^\top \beta_0 + b(\eta) + \epsilon \,. \end{aligned}$$

Consequently the distribution of $(Y(1), Y(0))$ is *not* independent of $T = \mathbb{1}(Q > 0)$ given X as both the responses and T are still correlated through η. Now we establish why it is still possible to identify α_0 in the presence of unobserved confounders and why the second equation of (1.2) is helpful. Note that for any realization (x, q) of (X, Q), we have:

$$\mathbb{E}[Y \mid Q = q, X = x] = \alpha_0 \mathbb{1}(q \geq 0) + x^\top \beta_0 + b(q - x^\top \gamma_0) \,.$$

Therefore, if b is a continuous function, we have:

$$\begin{aligned} &\lim_{h \downarrow 0} \{\mathbb{E}[Y \mid Q = h, X = x] - \mathbb{E}[Y \mid Q = -h, X = x]\} \\ &= \lim_{h \downarrow 0} \left\{\alpha_0 + b(h - x^\top \gamma_0) - b(-h - x^\top \gamma_0)\right\} = \alpha_0 \,. \end{aligned}$$

Therefore α_0 is identifiable from our model.

We, first, estimate γ_0 from $\mathcal{D}_1$ via standard least squares regression of Z on Q:

$$\hat{\gamma}_n = (Z^\top Z)^{-1} Z^\top Q\,.$$

Using $\hat{\gamma}_n$, we expand our first model equation as :

$$\begin{aligned} Y_i &= \alpha_0 S_i + X_i^\top \beta_0 + b(\hat{\eta}_i) + (\eta_i - \hat{\eta}_i) b'(\hat{\eta}_i) + R_{1,i} + \epsilon_i \\ &= \alpha_0 S_i + X_i^\top \beta_0 + b(\hat{\eta}_i) + b'(\hat{\eta}_i) Z_i^T (\hat{\gamma}_n - \gamma_0) + R_{1,i} + \epsilon_i\,, \end{aligned} \tag{2.2}$$

where $S_i = \mathbb{1}_{Q_i > 0}$, $\hat{\eta}_i = Q_i - Z_i^\top \hat{\gamma}_n$ and $R_{1,i} = (\hat{\eta}_i - \eta_i)^2 b''(\tilde{\eta}_i)/2$ is the residual obtained from a two-step Taylor expansion with $\tilde{\eta}_i$ lying between η_i and $\hat{\eta}_i$. It should be pointed out that if η_i were known, our model (equation (2.1)) would reduce to a simple partial linear model, and the estimation of α_0 would become straightforward. As we *do not* observe η_i, but rather use $\hat{\eta}_i$ as its proxy, the corresponding approximation error needs careful handling, as we need to show that the estimator does not inherit any resulting bias. Indeed, this is one of the core technical challenges of this paper and is explained in the subsequent development.

Note that the function b' in equation (2.2) is unknown. Since $\hat{\gamma}_n = \gamma_0 + O_p(n^{-1/2})$ (which, in turn, implies $\hat{\eta}_i = \eta_i + O_p(n^{-1/2})$), as long as we have a consistent estimate of b', say $\hat{b}'$, the approximation error $(b'(\hat{\eta}_i) - \hat{b}'(\hat{\eta}_i)) Z_i^\top (\hat{\gamma}_n - \gamma_0)$ is $o_p(n^{-1/2})$ and therefore asymptotically negligible.

We now elaborate on how we use a B-spline basis to estimate b' from $\mathcal{D}_2$. equation (2.1) can be rewritten as:

$$Y_i = \alpha_0 S_i + X_i^\top \beta_0 + b(\hat{\eta}_i) + \epsilon_i + R_i\,, \tag{2.3}$$

where $R_i = b(\eta_i) - b(\hat{\eta}_i)$. Ignoring the remainder term (which is shown to be asymptotically negligible under some mild smoothness condition on b to be specified later), we estimate b' via a B-spline basis from equation (2.3). The spline approximation theory is mostly explored for estimating compactly supported non-parametric regression functions. Although η is assumed to be unbounded, there are certain technical issues with estimating b' on the entire support (see Remark 2.2), and to circumvent that we restrict ourselves to a compact (but arbitrary) support $[-\tau, \tau]$, i.e., we consider those observations for which $|\hat{\eta}_i| \le \tau$ (this truncation is not necessary when η is bounded, e.g., in the scholarship example, η denotes some inherent skills which can be assumed to be bounded). We then use a cubic B-spline basis appropriately scaled to the interval of interest with equispaced knots to estimate b' [1]. Notationally, we use $K-1$ knots to divide $[-\tau, \tau]$ into K intervals of length $2\tau/K$ where $K = K_n$ increases with n at an appropriate rate (see remark 2.1 below), giving us in total $K+3$ spline basis functions. For any x, we use the notation $\tilde{N}_K(x) \in \mathbb{R}^{(K+3)}$ to denote the vector of scaled B-spline[2] basis functions $\{\tilde{N}_{K,j}\}_{j=1}^{K+3}$ evaluated at x. Using these basis functions we further expand equation (2.3) as follows:

$$Y_i = \alpha_0 S_i + X_i^\top \beta_0 + \tilde{N}_K(\hat{\eta}_i)^\top \omega_{b,\infty,n} + \epsilon_i + R_i + T_i\,, \tag{2.4}$$

for all those observations with $|\hat{\eta}_i| \le \tau$, where T_i is the spline approximation error, and $T_i = b(\hat{\eta}_i) - N_K(\hat{\eta}_i)^\top \omega_{b,\infty,n}$ and $\omega_{b,\infty,n}$ is the (population) parameter defined as:

$$\omega_{b,\infty,n} = \arg\min_{\omega \in \mathbb{R}^{(K+3)}} \sup_{|x| \le \tau} \left| b(x) - \tilde{N}_K^\top(x)\omega \right|\,.$$

[1] While we work with cubic splines, one may certainly use higher degree polynomials. However, in practice, it has been observed that cubic spline works well in most of the scenarios

[2] A brief discussion on B-spline basis and scaled B-spline basis is presented in Section 7 of Mukherjee, Banerjee and Ritov (2023) for the ease of the reader.

Suppose we have $n_2 \le n/3$ observations in $\mathcal{D}_2$ with $|\hat{\eta}_i| \le \tau$. Denote by $\mathbf{Y}$ the vector of all the corresponding n_2 responses, by $\mathbf{X} \in \mathbb{R}^{(n_2,p_1)}$ the covariate matrix, and by $\tilde{\mathbf{N}}_K \in \mathbb{R}^{(n_2,K+3)}$ the approximation matrix with rows $\tilde{\mathbf{N}}_{K,i*} = \tilde{N}_K(\hat{\eta}_i)$. Regressing Y on $(\mathbf{S}, \mathbf{X}, \tilde{\mathbf{N}}_K)$ we estimate $\omega_{b,\infty,n}$ (details can be found in the proof of Proposition 2.7) and set $\hat{b}'(x) = \nabla\tilde{N}_K(x)^\top \hat{\omega}_{b,\infty,n}$ where $\nabla\tilde{N}_K(x)$ is the vector of derivates of each of the co-ordinates of $\tilde{N}_K(x)$. The consistency of this estimator of b' is established in Proposition 2.7 in section 2.2.

Remark 2.1. *Henceforth, we choose $K \equiv K_n$ such that $n^{1/8} \ll K \ll n^{1/3}$ to control the approximation errors of certain non-parametric functions (including $b(\eta)$) that appear in our analysis via the B-spline basis. However, the bounds can be improved in the presence of additional derivations of the non-parametric functions involved.*

The final (key) step involves estimating α_0 using the observations in $\mathcal{D}_3$. Suppose there are $n_3 \sim n/3$ observations in $\mathcal{D}_3$ with $|\hat{\eta}_i| \le \tau$. Replacing b' by $\hat{b}'$ obtained from $\mathcal{D}_2$ in equation (2.2) we obtain:

$$\begin{aligned} Y_i &= \alpha_0 S_i + X_i^\top \beta_0 + b(\hat{\eta}_i) + \hat{b}'(\hat{\eta}_i) Z_i^\top (\hat{\gamma}_n - \gamma_0) + R_{1,i} + R_{2,i} + \epsilon_i \\ &\triangleq \alpha_0 S_i + X_i^\top \beta_0 + b(\hat{\eta}_i) + \tilde{Z}_i^\top (\hat{\gamma}_n - \gamma_0) + R_{1,i} + R_{2,i} + \epsilon_i \,, \\ Q_i - Z_i^\top \hat{\gamma}_n &= -Z_i^\top (\hat{\gamma}_n - \gamma_0) + \eta_i \,, \end{aligned} \tag{2.5}$$

where we define $\tilde{Z}_i = \hat{b}'(\hat{\eta}_i) Z_i$, the residual term $R_{2,i}$ as:

$$R_{2,i} = \left(b'(\hat{\eta}_i) - \hat{b}'(\hat{\eta}_i)\right) Z_i^\top (\hat{\gamma}_n - \gamma_0) = \left(b'(\hat{\eta}_i) - \hat{b}'(\hat{\eta}_i)\right)(\hat{\eta}_i - \eta_i) \,,$$

and $R_{1,i}$ is same as in equation (2.2). An inspection of the first part of equation (2.5) shows that up to the remainder terms $\left\{(R_{1,i}, R_{2,i})\right\}_{i=1}^{n_3}$, our model is a partial linear model with parameters (α_0, β_0, b). These remainder terms are asymptotically negligible as shown in the proof of our main theorem (Theorem 2.8). We estimate α_0 from equation (2.5) using standard techniques for the partial linear model, which, again, involve approximating the function b with the same B-spline basis as before:

$$\begin{aligned} Y_i &= \alpha_0 S_i + X_i^\top \beta_0 + \tilde{Z}_i^\top (\hat{\gamma}_n - \gamma_0) + b(\hat{\eta}_i) + R_{1,i} + R_{2,i} + \epsilon_i \\ &= \alpha_0 S_i + X_i^\top \beta_0 + \tilde{Z}_i^\top (\hat{\gamma}_n - \gamma_0) + \tilde{N}_K(\hat{\eta}_i)^\top \omega_{b,\infty,n} + R_{1,i} + R_{2,i} + R_{3,i} + \epsilon_i \,, \end{aligned} \tag{2.6}$$

where $R_{3,i}$ is the spline approximation error, i.e. $R_{3,i} = b(\hat{\eta}_i) - \tilde{N}_K(\hat{\eta}_i)^\top \omega_{b,\infty,n}$. Note that we estimate the conditional mean function $b(\cdot)$ twice here: once while estimating b' from $\mathcal{D}_2$ and again in the above equation. This second re-estimation of b is quite critical (i.e., we cannot *use* the estimator of b obtained from $\mathcal{D}_2$) due to the presence of higher-order bias (slower than $n^{-1/2}$). Combining equation (2.6) along with the second equation of (2.5) we formulate the following linear model equation:

$$\begin{aligned} \begin{pmatrix} \mathbf{Y} \\ \mathbf{Q} - \mathbf{Z}\hat{\gamma}_n \end{pmatrix} &= \begin{pmatrix} \mathbf{S} & \mathbf{X} & \tilde{\mathbf{Z}} & \tilde{\mathbf{N}}_K \\ 0 & 0 & -\mathbf{Z} & 0 \end{pmatrix} \begin{pmatrix} \alpha_0 \\ \beta_0 \\ \hat{\gamma}_n - \gamma_0 \\ \omega_{b,\infty,n} \end{pmatrix} + \begin{pmatrix} \mathbf{R} \\ 0 \end{pmatrix} + \begin{pmatrix} \epsilon \\ \eta \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{W} & \tilde{\mathbf{N}}_{K,a} \end{pmatrix} \begin{pmatrix} \theta_0 \\ \omega_{b,\infty,n} \end{pmatrix} + \begin{pmatrix} \mathbf{R} \\ 0 \end{pmatrix} + \begin{pmatrix} \epsilon \\ \eta \end{pmatrix} , \end{aligned} \tag{2.7}$$

where $\theta_0 = (\alpha_0, \beta_0, \hat{\gamma}_n - \gamma_0)$,

$$\mathbf{W} = \begin{bmatrix} \mathbf{W}_1 \\ \mathbf{W}_2 \end{bmatrix} \quad \text{with} \quad \mathbf{W}_1 = \begin{pmatrix} \mathbf{S} & \mathbf{X} & \tilde{\mathbf{Z}} \end{pmatrix}, \quad \mathbf{W}_2 = \begin{pmatrix} 0 & 0 & -\mathbf{Z} \end{pmatrix},$$

where $\mathbf{W}_1 \in \mathbb{R}^{(n_3, 1+p_1+p_2)}, \mathbf{W}_2 \in \mathbb{R}^{(n/3, 1+p_1+p_2)}$ (as we are using all the observations in $\mathcal{D}_3$ in the second regression equation of (2.5)), $\mathbf{R}$ is the vector of the sum of three residuals $R_{1,i}, R_{2,i}, R_{3,i}$ mentioned in equation (2.6), and the matrix $\tilde{\mathbf{N}}_{K,a}$ (read $\tilde{\mathbf{N}}_K$ appended) has the form:

$$\tilde{\mathbf{N}}_{K,a} = \begin{bmatrix} \tilde{\mathbf{N}}_K \\ 0 \end{bmatrix}.$$

From equation (2.7), we estimate θ_0 via ordinary least squares methods (an application of Frisch Waugh Lovell theorem):

$$\hat{\theta} = \left(\mathbf{W}^\top \mathsf{proj}^\perp_{\tilde{\mathbf{N}}_{K,a}} \mathbf{W} \right)^{-1} \mathbf{W}^\top \mathsf{proj}^\perp_{\tilde{\mathbf{N}}_{K,a}} \begin{pmatrix} \mathbf{Y} \\ \mathbf{Q} - \mathbf{Z}\hat{\gamma}_n \end{pmatrix},$$

and set $\hat{\alpha}$ as the first co-ordinate of $\hat{\theta}$, where for any matrix A, we define proj_A as the projection matrix on the columns space of A. Note that because of data-splitting, $(\hat{\gamma}_n, \hat{b}')$ is independent of $\mathcal{D}_3$, which provides a significant technical advantage in dealing with the asymptotics of our estimator.

Finally, we apply the same methodology on permutations of the three sets of data: i.e. we estimate $\hat{\gamma}_n$ from $\mathcal{D}_2$, $\hat{b}'$ from $\mathcal{D}_3$, $\hat{\alpha}$ from $\mathcal{D}_1$ and $\hat{\gamma}_n$ from $\mathcal{D}_3$, $\hat{b}'$ from $\mathcal{D}_1$, $\hat{\alpha}$ from $\mathcal{D}_2$. Denote by $\hat{\alpha}_i$, the estimator $\hat{\alpha}$ estimated from $\mathcal{D}_i$ for $1 \le i \le 3$. Our final estimate is then $\overline{\hat{\alpha}} = (1/3) \sum_i \alpha_i$. In literature, this method is also known as cross-fitting (see Chernozhukov et al. (2018)).

Remark 2.2. *As described in this section, we only use observations for which $|\hat{\eta}_i| \le \tau$, losing some efficiency in the process. One way to circumvent this issue is to use a sequence $\{\tau_n\}$ slowly increasing to ∞ and considering all those observations for which $|\hat{\eta}| \le \tau_n$. Although this will acquire efficiency in the limit, we need a stronger set of assumptions to make it work: for starters, we need conditions on the decay of the density of η; we need it not to vanish anywhere in $[-\tau_n, \tau_n]$, and knowledge of the rate at which $\min_{|x| \le \tau_n} f_\eta(x)$ approaches 0. We also need stronger conditions on some conditional expectation functions (i.e. conditional expectation of $\begin{pmatrix} S & \mathbf{X} & \mathbf{Z} \end{pmatrix}$ given $\eta + a^\top Z$ for some vector a), e.g., bounded derivatives in both co-ordinates over the entire space (see Lemma 2.1 of Mukherjee, Banerjee and Ritov (2023)). With more technical nuances, we believe our method can be extended to the entire real line by using a growing interval, but from a purely statistical angle, it will not bring anything insightful to the methodology that we propose here.*

2.2. Theoretical analysis

In this section, we present our main theorems with a broad outline of the proofs. Details are provided in the Mukherjee, Banerjee and Ritov (2023). To establish the theory, we need the following assumptions:

Assumption 2.3. *The distribution of (η, ν) satisfies the following conditions:*

1. *The density of η, denoted by f_η, is continuously differentiable and both f_η and its derivatives are uniformly bounded.*
2. *The conditional mean function $b(\eta) = \mathbb{E}[\nu \mid \eta]$ is 3 times differentiable and b'' and b''' are uniformly bounded over real line.*

3. *The variance function* $\sigma^2(\eta) = \mathsf{var}(\epsilon \mid \eta)$ *is uniformly bounded from above.*
4. *There exists some* $\xi > 0$ *such that:* $\min_{|x| \le \tau + \xi} f_\eta(x) > 0$.

Assumption 2.4. *Define the matrices* Ω *and* Ω^* *as:*

$$\Omega = \mathbb{E}\left[\mathsf{cov}\left(\begin{bmatrix} S & X & Zb'(\eta) \end{bmatrix} \mid \eta\right)\right] + \mathsf{cov}\left(\begin{bmatrix} 0 & 0 & Z \end{bmatrix}\right),$$
$$\Omega^* = \mathbb{E}\left[\sigma^2(\eta)\mathsf{cov}\left(\begin{bmatrix} S & X & Zb'(\eta) \end{bmatrix} \mid \eta\right)\right] + \mathsf{var}(\eta)\mathsf{cov}\left(\begin{bmatrix} 0 & 0 & Z \end{bmatrix}\right).$$

Then, the minimum eigenvalues of Ω *and* Ω^* *are strictly positive.*

Assumption 2.5. *The distribution of* (X, Z) *satisfies the following conditions:*

1. (X, Z) *has a bounded continuous density function and has zero expectation.*
2. *The first four moments of* X, Z *are finite.*

Remark 2.6. *Assumption 2.3 provides a low-level assumption on the smoothness of the density of* η*, the conditional mean function* $b(\eta)$ *and the conditional variance profile* $\sigma^2(\eta)$*, which is required for the standard asymptotic analysis of the partial linear model. Assumption 2.4, again, is a standard assumption in partial linear model literature. It is essential for the asymptotic normality of the treatment effect as the asymptotic variance of our estimator is a function of these variances. If this assumption is violated, then the asymptotic variance will be infinite, and that estimation at* $\sqrt{n}$ *is not possible. (Note that if* $\gamma_0 = 0$*, then Assumption 2.4 is violated.) As our method does not use all observations but a fraction depending on the interval* $[-\tau, \tau]$*, our limiting variance comprises of the following truncated versions of* Ω *and* Ω^**:*

$$\Omega_\tau = \mathbb{E}\left[\mathsf{cov}\left(\begin{bmatrix} S & X & Zb'(\eta) \end{bmatrix} \mid \eta\right) \mathbb{1}_{|\eta| \le \tau}\right] + \mathsf{cov}\left(\begin{bmatrix} 0 & 0 & Z \end{bmatrix}\right),$$
$$\Omega^*_\tau = \mathbb{E}\left[\sigma^2(\eta)\mathsf{cov}\left(\begin{bmatrix} S & X & Zb'(\eta) \end{bmatrix} \mid \eta\right) \mathbb{1}_{|\eta| \le \tau}\right] + \mathsf{var}(\eta)\mathsf{cov}\left(\begin{bmatrix} 0 & 0 & Z \end{bmatrix}\right).$$

It is immediate that if $\tau \to \infty$*, then* $\Omega_\tau \to \Omega$ *and* $\Omega^*_\tau \to \Omega^*$*. Hence, in light of Assumption 2.4, by continuity, the minimum eigenvalues of* Ω_τ *and* Ω^*_τ *are also positive for all large* τ*.*

2.2.1. Asymptotic normality

The following theorem establishes consistency of our estimator of b' (proof can be found in Section 4 of the Mukherjee, Banerjee and Ritov (2023)):

Proposition 2.7. *Under Assumptions 1.1, 2.4 - 2.5, we have:*

$$\sup_{|x| \le \tau} \left|b'(x) - \hat{b}'(x)\right| = \sup_{|x| \le \tau} \left|b'(x) - \nabla \tilde{N}_K(x)^\top \hat{\omega}_{b,\infty,n}\right| = o_p(1).$$

We now state the main result of the paper. A proof sketch highlighting the main argument is presented in Section 1.1 of Mukherjee, Banerjee and Ritov (2023), and the detailed proof can be found in Section 2 of Mukherjee, Banerjee and Ritov (2023).

Theorem 2.8. *Consider the estimates obtained at the end of the previous section. Under Assumptions 1.1, 2.4-2.5:*

$$\sqrt{n}\,(\hat{\alpha}-\alpha_0) \overset{\mathscr{L}}{\Longrightarrow} \mathcal{N}(0, 3e_1^\top \Omega_\tau^{-1}\Omega_\tau^{*}\Omega_\tau^{-1} e_1)\,,$$

whilst

$$\sqrt{n}\,(\bar{\hat{\alpha}}-\alpha_0) \overset{\mathscr{L}}{\Longrightarrow} \mathcal{N}(0, e_1^\top \Omega_\tau^{-1}\Omega_\tau^{*}\Omega_\tau^{-1} e_1)\,,$$

Here, e_1 is the vector with the first coordinate 1, and the other coordinates are 0.

Remark 2.9. *For inference, we need a consistent estimate of the asymptotic variance, namely $\Omega_\tau^{-1}\Omega_\tau^{*}\Omega_\tau^{-1}$. A consistent estimator of Ω_τ can be obtained immediately from Step 1. However, estimation of $\Omega_\tau^{\star}$ is hard as it is the variance of the efficient influence function $\varphi(X, Z, \nu, \eta)$ (modulo the truncation). The expression for the efficient influence function is:*

$$\begin{aligned}&\varphi(X_i, Z_i, \eta_i, \nu_i)\\ &= \left\{\begin{bmatrix} S_i & X_i & b'(\eta_i)Z_i\end{bmatrix} - \mathbb{E}\left[\begin{bmatrix} S_i & X_i & b'(\eta_i)Z_i\end{bmatrix}\middle|\eta=\eta_i\right]\right\}\epsilon_i \mathbb{1}_{|\eta_i|\le\tau} + \begin{bmatrix} 0 & 0 & Z_i\eta_i\end{bmatrix}.\end{aligned}$$

One possible way to estimate the variance is to replace b', η, and ϵ by their approximations and then calculate the variance, but typically it will be quite cumbersome in practice. Hence, we believe it is better to use other techniques like bootstrap for variance estimation (see Section 6.2).

2.2.2. Semi-parametric efficiency

We further show that our estimator is semi-parametrically efficient under certain restrictions. As our estimator is based on the least square approach, it can not be shown to be efficient unless the error ϵ is normal. We prove the following theorem (proof can be found in Section 3.2 of Mukherjee, Banerjee and Ritov (2023)):

Theorem 2.10. *Suppose the model is the following:*

$$Y_i = \alpha_0 \mathbb{1}_{Z_i^\top\gamma_0+\eta>0} + X_i^\top\beta_0 + b(\eta_i) + \epsilon_i\,,$$

where $\epsilon_i \sim \mathcal{N}(0,\tau^2) \perp\!\!\!\perp \eta_i$. Then our estimator of α_0 is semi-parametrically efficient, i.e. its asymptotic variance $\sigma_0^2 = \tau^2 e_1^\top \Omega^{-1} e_1$ attains the semi-parametric information bound for this model.

Remark 2.11. *The assumption of the normality of ϵ is necessary to establish semiparametric efficiency for least squares type methods, but the assumption of homoskedasticity is essential only if we use the ordinary least squares method. One may easily take care of heteroskedasticity by using weighted least squares instead. The first step towards that direction is to approximate the variance profile $\sigma(\eta)$ using $\hat{\sigma}(\hat{\eta})$ for some non-parametric estimate $\hat{\sigma}(\cdot)$ of $\sigma(\cdot)$. Then defining $\mathbf{D} \in \mathbb{R}^{(n_3+n/3)\times(n_3+n/3)}$ to be the diagonal matrix with first $n/3$ diagonal entries being $\hat{\sigma}(\hat{\eta}_i)$'s (i.e. for all those $\hat{\eta}_i$'s such that $|\hat{\eta}_i| \le \tau$) and last $n/3$ diagonal entries being $\hat{\sigma}_\eta$'s (an estimate of variance of η) we estimate the treatment effect as:*

$$\hat{\alpha} = e_1^\top \left(\mathbf{W}^\top \mathbf{D}^{-1/2}\mathsf{proj}^{\perp}_{\mathbf{D}^{-1/2}\tilde{\mathbf{N}}_{K,a}}\mathbf{D}^{-1/2}\mathbf{W}\right)^{-1}\mathbf{W}^\top\mathbf{D}^{-1/2}\mathsf{proj}^{\perp}_{\mathbf{D}^{-1/2}\tilde{\mathbf{N}}_{K,a}}\mathbf{D}^{-1/2}\begin{pmatrix}\mathbf{Y}\\ \mathbf{Q}-\mathbf{Z}\hat{\gamma}_n\end{pmatrix}.$$

A more tedious analysis establishes that this estimator is asymptotic normal and semi-parametrically efficient under the error structure: $\nu_i = b(\eta_i) + \sigma(\eta_i)\epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0,1)$. As this does not add

anything of significance to the core idea of the paper, we confine ourselves to using OLS instead of WLS for ease of presentation.

3. Estimation procedure for α_0 in high dimension

As highlighted in our analysis for fixed dimensional covariates in the previous section, our model is an example of a partial linear model, where the observations corresponding to the non-linear part are noisy, and the noise is correlated with the covariates corresponding to the linear part. Recall that our is model is defined as:

$$\begin{aligned} Y_i &= \alpha_0 \mathbb{1}_{Q_i \geq 0} + X_i^\top \beta_0 + b(\eta_i) + \epsilon_i \,, \\ Q_i &= Z_i^\top \gamma_0 + \eta_i \,, \end{aligned} \tag{3.1}$$

with $\mathbb{E}[\eta \mid Z] = 0$ and $\mathbb{E}[\epsilon \mid \eta, X, Z] = 0$. In this section, we assume that the dimension of (X, Z) is larger than the sample size. More specifically, denoting $p_1 := \dim(X)$ and $p_2 := \dim(Z)$, we assume that $\min\{p_1, p_2\} \gg n$. As before, we only observe (Y_i, X_i, Z_i, Q_i) but not η_i. As in the case for almost all high dimensional statistical analysis, we assume that both β_0 and γ_0 are sparse vectors (with $\|\beta_0\|_0 = s_\beta$ and $\|\gamma_0\|_0 = s_\gamma$ where $\max\{s_\beta, s_\gamma\} \ll n$) to enable consistent estimation of the treatment effect α_0. We will quantify the precise assumptions needed for our theory later. The core estimation procedure in this high-dimensional regime is similar to its fixed-dimensional counterpart but with certain changes to account for the effect of high-dimensional covariates. First, we highlight the key changes here, and then we elaborate on those changes.

1. As before, we divide the whole data $\mathcal{D}$ into three parts $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$ (with n_i data in $\mathcal{D}_i$, where $n_1 \sim n_2 \sim n_3 \sim n/3$). From $\mathcal{D}_1$, we estimate γ_0 by performing LASSO regression instead of OLS of Q on Z to take care of the high dimensionality of Z:

$$\hat{\gamma} = \arg\min_{\gamma \in \mathbb{R}^{p_2}} \left[\frac{1}{2n_3} \|\mathbf{Q} - \mathbf{Z}\gamma\|^2 + \lambda \|\gamma\|_1 \right] ,$$

   where $\lambda \asymp \sqrt{\log p_2 / n}$ and set $\hat{\eta}_i = Q_i - Z_i^\top \hat{\gamma}$ for all observations in $\mathcal{D}_2 \cup \mathcal{D}_3$.
2. To estimate b' (using the observations in $\mathcal{D}_2$) we expand the first equation of (3.1) as:

$$Y_i = \alpha_0 \mathbb{1}_{Q_i \geq 0} + X_i^\top \beta_0 + b(\hat{\eta}_i) + R_i + \epsilon_i \,. \tag{3.2}$$

   Here, we use techniques for the high dimensional partial linear model (as now X is high-dimensional, e.g., see Müller and Van de Geer (2015)) to estimate b', which is quite different and technically involved than the one used in the analysis for fixed dimensional covariates.
3. Upon estimating b', we obtain the following equation similar to (2.6):

$$Y_i = \alpha_0 \mathbb{1}_{Q_i \geq 0} + X_i^\top \beta_0 + \tilde{\mathbf{N}}_k(\hat{\eta}_i)^\top \omega_b + Z_i^\top (\hat{\gamma}_n - \gamma_0) \hat{b}'(\hat{\eta}_i) + \tilde{R}_i + \epsilon_i \,, \tag{3.3}$$

   where we use techniques related to the analysis of the debiased LASSO estimator for estimating α_0.

In the first step, we move from OLS to LASSO as Z is high dimensional. We show that the estimator of γ_0 obtained in the first step is consistent and rate optimal under the restricted eigenvalue assumption (henceforth RE) on $\mathbf{Z}$ (e.g., see Assumption RE of Bickel et al. (2009)). Just like in our analysis for fixed dimension, we only consider all the $\hat{\eta}$'s (both in $\mathcal{D}_2$ and $\mathcal{D}_3$) such that $|\hat{\eta}| \leq \tau$ and ignore the

remaining data from $\mathcal{D}_2$ and $\mathcal{D}_3$. Therefore, the rest of our analysis on $\mathcal{D}_2$ and $\mathcal{D}_3$ are solely based on those observations for which $|\hat{\eta}| \leq \tau$. The second step is obtained via replacing $b(\eta_i)$ by $b(\hat{\eta}_i)$ for the observations in $\mathcal{D}_2$ with the residuals being $R_i = b(\eta_i) - \hat{b}(\eta_i)$. However, as X is high-dimensional, we have to invoke techniques from the analysis of high-dimensional partial linear models to estimate b' using the above equation, which we describe in detail subsequently. The last step follows from a two-step Taylor expansion as before; replacing $b'(\hat{\eta}_i)$ by $\hat{b}'(\hat{\eta}_i)$, we obtain the following representation for the observations in $\mathcal{D}_3$:

$$\begin{aligned} Y_i = \alpha_0 \mathbb{1}_{Q_i \geq 0} + X_i^\top \beta_0 + b(\hat{\eta}_i) + Z_i^\top (\hat{\gamma}_n - \gamma_0) \hat{b}'(\hat{\eta}_i) \\ + Z_i^\top (\hat{\gamma}_n - \gamma_0) \left(b'(\hat{\eta}_i) - \hat{b}'(\hat{\eta}_i) \right) + (\eta_i - \hat{\eta}_i)^2 b''(\tilde{\eta}_i) + \epsilon_i \,. \end{aligned}$$

Further, an approximation of $b(\hat{\eta}_i)$ by B-spline basis yields:

$$Y_i = \alpha_0 \mathbb{1}_{Q_i \geq 0} + X_i^\top \beta_0 + \tilde{\mathbf{N}}_k(\hat{\eta}_i)^\top \omega_b + Z_i^\top (\hat{\gamma}_n - \gamma_0) \hat{b}'(\hat{\eta}_i) + \tilde{R}_i + \epsilon_i \,, \tag{3.4}$$

where $\mathbf{N}_k(\eta)$ consists of the B-spline basis functions evaluated at $\hat{\eta}_i$. The residual term $\tilde{R}_i$ can be decomposed as $R_{1,i} + R_{2,i} + R_{3,i}$ where the individual residuals are:

$$\begin{aligned} R_{1,i} &= b(\hat{\eta}_i) - \tilde{\mathbf{N}}_k(\hat{\eta}_i)^\top \omega_b \,, \\ R_{2,i} &= Z_i^\top (\hat{\gamma}_n - \gamma_0) \left(b'(\hat{\eta}_i) - \hat{b}'(\hat{\eta}_i) \right) \,, \\ R_{3,i} &= (\eta_i - \hat{\eta}_i)^2 b''(\tilde{\eta}_i) \,. \end{aligned}$$

The first residual is the B-spline approximation error of b while the second is the product of two different error terms: (i) the error in the estimation of γ_0 from $\mathcal{D}_1$ and (ii) the error in the estimation of b' from $\mathcal{D}_2$ and the last is the Taylor approximation error. We use equation (3.4) as our main estimating equation for α_0. Before delving into the estimation procedure, we introduce some notation:

1. $\check{Z}_i$ will be used to denote $Z_i \hat{b}'(\hat{\eta}_i)$.
2. The vector W will be used to denote the random vector $(S, X, Zb'(\eta))$.
3. $\check{W}$ will be used to denote the random vector $(S, X, \check{Z})$ and the jth element of W (resp. $\check{W}$) will be denoted as W_j (resp. $\check{W}_j$).
4. $m_j(\eta)$ will be used to denote $\mathbb{E}[W_j \mid \eta]$, for all $0 \leq j \leq 1 + p_1 + p_2$ where $W_0 = Y$.
5. $\check{m}_j(\hat{\eta})$ will be used to denote $\mathbb{E}[W_j \mid \hat{\eta}]$, for all $0 \leq j \leq 1 + p_1 + p_2$ where $W_0 = Y$.
6. $\widehat{\tilde{W}}$ will be used to denote $W - \mathbb{E}[W \mid \hat{\eta}]$.
7. $\tilde{W}$ will be used to denote the random vector $W - \mathbb{E}[W \mid \eta]$.
8. $\check{\mathbf{W}}, \hat{\tilde{\mathbf{W}}}, \tilde{\mathbf{W}}$ will be used to denote the matrix version of $\check{W}, \hat{\tilde{W}}, \tilde{W}$ respectively by concatenating all of the n_3 observations of $\mathcal{D}_3$ row-wise. Similarly, $\mathbf{S}$ and $\tilde{\mathbf{S}}$ denote the vectorized version of S and $\tilde{S} = S - \mathbb{E}[S \mid \eta]$. Note that $\mathbf{S}$ (resp. $\tilde{\mathbf{S}}$) is the first column of $\mathbf{W}$ (resp. $\tilde{\mathbf{W}}$).
9. For any matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ and for any $1 \leq j \leq d$, we denote by $\mathbf{X}_{-j}$ the matrix which is obtained by removing the jth column of $\mathbf{X}$. Similarly for any matrix Σ, we denote by $\Sigma_{-i,-j}$ to be the matrix obtained via removing i^{th} row and j^{th} column of Σ.
10. $\theta_0 \equiv \theta_{0,n} = (\alpha_0, \beta_0, \hat{\gamma}_n - \gamma_0)$.
11. $\tilde{\mathbf{N}}_k$ will be used to denote the $n \times K$ matrix, whose i^{th} row consists of K B-spline bases evaluated at $\hat{\eta}_i$.
12. For any vector v (or matrix $\mathbf{A}$), the notation $v^\perp$ (resp. $\mathbf{A}^\perp$) is used to denote $P^\perp_{\tilde{\mathbf{N}}_k} v$ (resp. $P^\perp_{\tilde{\mathbf{N}}_k} A$).

Using the above notation, equation (3.4) can be rewritten (in matrix form concatenating all the observations along the rows) as:

$$\mathbf{Y} = \breve{\mathbf{W}}\theta_0 + \mathbf{N}_k\omega_b + \tilde{\mathbf{R}} + \boldsymbol{\epsilon}\,.$$

Projecting out the effect of $\mathbf{N}_k$ from both sides yields:

$$P^{\perp}_{\tilde{\mathbf{N}}_k}\mathbf{Y} = P^{\perp}_{\tilde{\mathbf{N}}_k}\breve{\mathbf{W}}\theta_0 + P^{\perp}_{\tilde{\mathbf{N}}_k}\tilde{\mathbf{R}} + P^{\perp}_{\tilde{\mathbf{N}}_k}\boldsymbol{\epsilon}\,.$$

Our estimate of α_0 is defined as follows:

$$\hat{\alpha} = \frac{\left(\mathbf{Y}^{\perp} - \breve{\mathbf{W}}^{\perp}_{-1}\hat{\theta}_{-1,Y}\right)^{\top}\left(\mathbf{S}^{\perp} - \breve{\mathbf{W}}^{\perp}_{-1}\hat{\theta}_{-1,S}\right)}{\left(\mathbf{S}^{\perp} - \breve{\mathbf{W}}^{\perp}_{-1}\hat{\theta}_{-1,S}\right)^{\top}\left(\mathbf{S}^{\perp} - \breve{\mathbf{W}}^{\perp}_{-1}\hat{\theta}_{-1,S}\right)}\,,$$

where $\hat{\theta}_{-1,Y}$ denotes the LASSO estimator obtained by regressing $\mathbf{Y}$ on $\breve{\mathbf{W}}_{-1}$, i.e.

$$\hat{\theta}_{-1,Y} = \arg\min_{\theta}\left[\frac{1}{2n_3}\left\|\mathbf{Y} - \breve{\mathbf{W}}_{-1}\theta\right\|^2 + \lambda_0\left\|\theta\right\|_1\right]\,, \tag{3.5}$$

and $\hat{\theta}_{-1,S}$ denotes the LASSO estimator obtained by regressing $\mathbf{S}$ on $\breve{\mathbf{W}}_{-1}$, i.e.

$$\hat{\theta}_{-1,S} = \arg\min_{\theta}\left[\frac{1}{2n_3}\left\|\mathbf{S} - \breve{\mathbf{W}}_{-1}\theta\right\|^2 + \lambda_1\left\|\theta\right\|_1\right]\,, \tag{3.6}$$

for some appropriate choice of λ_0, λ_1 to be specified later. Here n_3 is the number of observations in $\mathcal{D}_3$ for which $|\hat{\eta}_i| \le \tau$. It is immediate that $n_3 \asymp n/3 \asymp n$, i.e., the number of observations in $\mathcal{D}_3$ with $|\hat{\eta}_i| \le \tau$ is of the order n, the total number of observations. Henceforth, we will ignore this difference and state all our results in terms of n. We show that $\sqrt{n_3}\,(\hat{\alpha} - \alpha_0)$ is asymptotically normal under certain assumptions, which are stated below:

Assumption 3.1 (Mean independence). *The errors (ϵ, η) are assumed to satisfy $\mathbb{E}[\epsilon \mid \eta, X, Z] = 0$ and $\mathbb{E}[\eta \mid Z] = 0$ and the conditional variances $\mathsf{var}(\epsilon \mid X, Z, \eta)$ and $\mathsf{var}(\eta \mid Z)$ are uniformly bounded from above.*

Remark 3.2. *Assumption 3.1 is weaker than Assumption 1.1 for fixed-dimensional covariates. However, with additional technical manipulations, our analysis can be extended to the mean independence case for the fixed-dimensional scenario. Our aim there was to keep the model simple and illustrate the basic ideas of the analysis. In this high-dimensional setup, we invoke the weak mean independence assumption and present the technicalities necessary to establish the asymptotic normality of the estimator under this more general scenario.*

Assumption 3.3 (Smoothness of b). *The function b in equation (3.1) is assumed to be $\upsilon \ge 3$ times differentiable with bounded derivates.*

Assumption 3.4 (Lower bound on the density of η). *Same as part (4) of Assumption 2.3.*

Assumption 3.5 (Smoothness of conditional expectation). *Define the function $g_j(a,t)$ as:*

$$g_j(a,t) = \mathbb{E}\left[W_j \mid \eta + a^{\top}Z = t\right]\,,$$

for $0 \le j \le 1 + p_1 + p_2$. Assume for any fixed $\|a\| \le r$, the collection $\{g_j(a,\cdot)\}_{1\le j\le p}$ belongs to a function class $\Sigma(\alpha, \mathbf{l})$ for some $\alpha \ge 3$ and $\mathbf{l} \in \mathbb{R}^{\lfloor\alpha\rfloor}$, where $\Sigma(\alpha,\mathbf{l})$ is defined as the collection of all the functions f, which are $\lfloor\alpha\rfloor$ times differentiable and the $\lfloor\alpha\rfloor^{th}$ derivative satisfies:

$$\left|f^{\lfloor\alpha\rfloor}(x) - f^{\lfloor\alpha\rfloor}(y)\right| \le \mathbf{l}_{\lfloor\alpha\rfloor}\,|x-y|^{\alpha-\lfloor\alpha\rfloor}\,,$$

and $\|f^{(i)}\|_\infty \le \mathbf{l}_i$ for all $1 \le i \le \lfloor\alpha\rfloor - 1$. Moreover, this $\mathbf{l}$ is uniform over all $\|a\| \le r$ where r is independent of the underlying dimension.

Remark 3.6. *The conditional mean functions m_j and $\check{m}_j$ are special cases of g_j as $\check{m}_j(t) = g_j(\hat{\gamma}_n - \gamma_0, t)$ and $m_j(t) = g_j(0,t)$.*

Assumption 3.7 (Sub-gaussianity). *Assume that for $0 \le j \le p_1 + p_2$, $\tilde{W}_j, \hat{\tilde{W}}_j, m_j(\eta), \check{m}_j(\hat{\eta}), \eta$ and ϵ are subgaussian random-variables with subgaussianity constant uniformly bounded by σ_W. Furthermore assume that $\mathsf{var}(W_j \mid \hat{\eta})$ is uniformly (over j) bounded on $[-\tau,\tau]$.*

For our next assumption, we define a covariance matrix Σ and two vectors $\theta_Y^* \in \mathbb{R}^{p_1+p_2}$ and $\theta_S^* \in \mathbb{R}^{p_1+p_2}$ which play a crucial role in our analysis:

$$\begin{aligned}
\Sigma_\tau &= \mathbb{E}_\eta\left[(W - \mathbb{E}[W \mid \eta])\,(W - \mathbb{E}[W \mid \eta])^\top \mathbb{1}_{|\eta|\le\tau}\right] \\
&= \mathbb{E}[\tilde{W}\tilde{W}^\top \mathbb{1}_{|\eta|\le\tau}] \\
&= \mathbb{E}_\eta\left[\mathsf{var}(W \mid \eta)\mathbb{1}_{|\eta|\le\tau}\right] \\
&= \mathbb{E}_\eta\left[\mathsf{var}\left(\begin{pmatrix} S & X & b'(\eta)Z\end{pmatrix} \mid \eta\right)\mathbb{1}_{|\eta|\le\tau}\right],
\end{aligned}$$

$$\theta_S^* = \arg\min_\delta \mathbb{E}\left[\left(\tilde{S} - \tilde{W}_{-1}^\top\delta\right)^2 \mathbb{1}_{|\eta|\le\tau}\right] = \left(\Sigma_{\tau_{-1,-1}}\right)^{-1}\mathbb{E}[\tilde{W}_{-1}\tilde{S}\mathbb{1}_{|\eta|\le\tau}]\,, \tag{3.7}$$

$$\theta_Y^* = \arg\min_\delta \mathbb{E}\left[\left(\tilde{Y} - \tilde{W}_{-1}^\top\delta\right)^2 \mathbb{1}_{|\eta|\le\tau}\right] = \left(\Sigma_{\tau_{-1,-1}}\right)^{-1}\mathbb{E}[\tilde{W}_{-1}\tilde{Y}\mathbb{1}_{|\eta|\le\tau}]\,. \tag{3.8}$$

The matrix Σ_τ is the expectation of conditional variance of W given η on the set $|\eta| \le \tau$, which arises in the estimation of the linear part of a partial linear model. This matrix can be thought of as a high dimensional analog of Ω_τ defined in our analysis for fixed dimensional covariates. θ_Y^* (resp. $\hat{\theta}_S^*$) is the best linear estimator in the population for regressing $\tilde{Y}$ (resp. $\tilde{S}$) on $\tilde{W}$. In the definition of our estimator $\hat{\alpha}$, we have regressed $\mathbf{Y}^\perp$ on $\check{\mathbf{W}}_{-1}^\perp$ and $\mathbf{S}^\perp$ on $\check{\mathbf{W}}_{-1}^\perp$. Intuitively speaking, projecting out the column space of $\mathbf{N}_k(\hat{\eta})$ is asymptotically equivalent to centering around the conditional expectation with respect to η. Therefore, it is expected that $\hat{\theta}_{-1,Y}$ (respectively $\hat{\theta}_{-1,S}$) should be asymptotically consistent for regressing θ_Y^* (respectively $\hat{\theta}_S^*$) under certain sparsity assumption and Restricted eigenvalue (RE) condition on the covariate matrix $\check{\mathbf{W}}_{-1}^\perp$. We now state our assumptions on $\Sigma, \theta_Y^*, \theta_S^*$:

Assumption 3.8 (Asymptotic variance). *Assume that there exist $C_{\min} > 0$ and $C_{\max} < \infty$ such that:*

$$C_{\min} \le \lambda_{\min}(\Sigma_\tau) \le \lambda_{\max}(\Sigma_\tau) \le C_{\max} < \infty\,.$$

Furthermore, define

$$\sigma_{n,1}^2 = \mathbb{E}\left[\epsilon^2\left(\tilde{S} - \tilde{W}_{-1}^\top\theta_S^*\right)^2 \mathbb{1}_{|\eta|\le\tau}\right].$$

Assume that:

$$C_{\min} \leq \liminf_n \sigma_{n,1}^2 \leq \limsup_n \sigma_{n,1}^2 \leq C_{\max} < \infty\,,$$

$$\limsup_{n\to\infty} \mathbb{E}\left[\left(\tilde{S} - \tilde{W}_{-1}^\top \theta_S^*\right)^{2+\xi}\right] < \infty\,,$$

for some small $\xi > 0$.

Assumption 3.9 (Sparsity). *Assume there exist* $s_0, s_1 > 0$ *such that* $\|\theta_Y^*\|_0 \leq s_0$, $\|\theta_S^*\|_0 \leq s_1$ *and* $s_i \log(p_1 \wedge p_2)/n \to 0$.

Remark 3.10. *Although it is apparent from equations (3.5) and (3.6) that RE condition is needed on* $\check{\mathbf{W}}_{-1}^{\perp}$ *for the consistency and rate optimality of LASSO estimates* $\hat{\theta}_{-1,S}$ *and* $\theta_{-1,1}^*$*, from the sub-gaussianity of* $\tilde{W}$*, we have that* $\tilde{\mathbf{W}}_{-1}$ *satisfies RE condition with high probability (see Rudelson and Zhou (2012)). We prove in Proposition 5.5 that this implies* $\check{\mathbf{W}}_{-1}^{\perp}$ *also satisfies RE condition with high probability and consequently* $\hat{\theta}_{-1,S}$ *and* $\hat{\theta}_{-1,Y}$ *will be rate efficient.*

Remark 3.11. *From our original model equation 3.1, we have:*

$$\tilde{Y} = Y - \mathbb{E}[Y \mid \eta] = \tilde{S}\alpha_0 + \tilde{X}\beta_0\,,$$

which implies

$$\theta_Y^* = \left(\Sigma_{\tau_{-1,-1}}\right)^{-1} \mathbb{E}[\tilde{W}_{-1}\tilde{Y}\mathbb{1}_{|\eta|\leq\tau}] = \theta_S^*\alpha_0 + \theta_{0,-1}^*\,,$$

where $\theta_{0,-1}^* = (\beta_0, 0)$*. Note that* $\theta_{0,-1}^*$ *is already sparse with* $\left\|\theta_{0,-1}^*\right\|_0 = s_\beta$ *(recall that we have defined* $s_\beta = \|\beta_0\|_0$*). So, sparsity assumption on* θ_S^* *automatically ensures sparsity of* θ_Y^**; in other words, we have* $s_0 \leq s_1 + s_\beta$.

Under the above assumptions, we now present our main theorem regarding the asymptotic normality of our estimator in the high-dimensional setup below. A roadmap of the proof highlighting the key steps is presented in Section 1.2, and the detailed proof can be found in Section 5 of Mukherjee, Banerjee and Ritov (2023).

Theorem 3.12. *Under Assumptions 3.1-3.5, 3.8, 3.9, and certain conditions on the sparsity parameters* $s_0, s_1, s_\beta, s_\gamma$ *(see subsection 5.4 of Mukherjee, Banerjee and Ritov (2023) for the details), we have:*

$$\frac{\sigma_{n,2}^2}{\sigma_{n,1}}\sqrt{n_3}\,(\hat{\alpha} - \alpha_0) \overset{\mathscr{L}}{\Longrightarrow} \mathcal{N}(0,1)\,,$$

where the values of $\sigma_{n,1}$ *is as defined in Assumption 3.8 and* $\sigma_{n,2}$ *is:*

$$\sigma_{n,2} = \sqrt{\mathbb{E}\left[\left(\tilde{S} - \tilde{W}_{-1}\theta_S^*\right)^2 \mathbb{1}_{|\eta|\leq\tau}\right]} = \frac{1}{\sqrt{\left(\Sigma_\tau^{-1}\right)_{1,1}}}\,.$$

Remark 3.13. *Similar to our analysis for the fixed dimensional covariates, we can gain efficiency (in terms of asymptotic variance) here as well by rotating the datasets and taking an average of the estimates of α_0). However, as the analysis is already quite involved, we do not pursue this extension here.*

Remark 3.14. *As we do not assume $X = Z$ (they may or may not have overlap), it may be possible that one of them is high-dimensional while the other is not. In that case, the problem will be significantly easier to solve than the one analyzed in this section. For example, if Z is low-dimensional, we can estimate η simply using residual from linear regression (without penalty) in Step 1. On the other hand, if X is low-dimensional, then the procedure in Step 3 will be exactly the same as that in Section 2, i.e., we will not use techniques akin to those employed for debiasing lasso estimators in high dimension.*

4. Simulation studies

In this section, we present some simulation results for fixed dimension, i.e., $\dim(X)$ is fixed) and growing dimensional, i.e. $\dim(X)$ grows with n. We present our findings in two subsections.

4.1. Simulation for fixed dimension

We set $p = 2$ and $n = 5000$ in this fixed-dimensional regime. The setup is as follows:

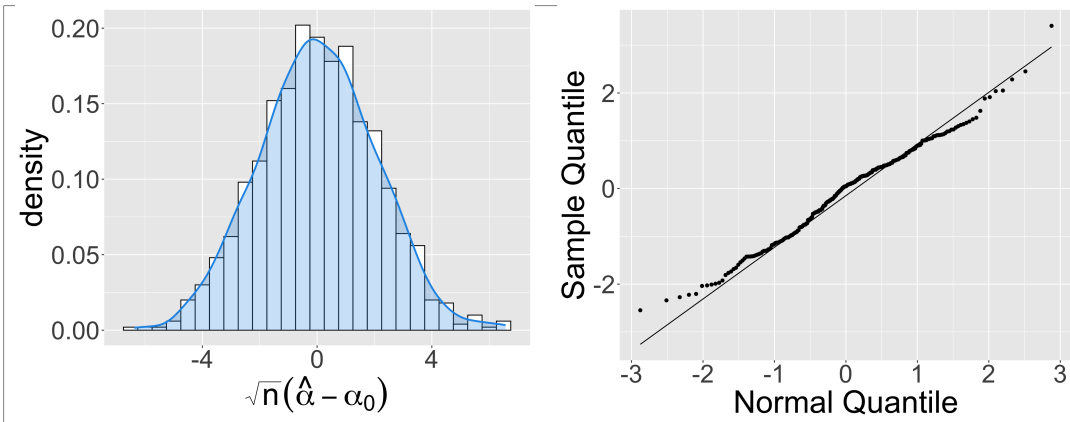
1. The true parameters are $\alpha_0 = 2$, $\beta_0 = (1, 2)$ and $\gamma_0 = (-1, 1)$ and $b(x) = x^3/3$.
2. The co-ordinates of the covariate X is generated independently from $\mathsf{unif}(-1, 1)$.
3. The errors (η, ϵ) are generated independently from $\mathsf{unif}(-1, 1)$.
4. We set $Z = X$ and $\tau_0 = 0$ to generate Q.

Using this data generation mechanism, we run our algorithms 500 times (Monte Carlo iterations) to estimate α_0. The results are presented in Figure 1. Subfigure (a) displays the histogram of $\sqrt{n}(\hat{\alpha} - \alpha_0)$, with the curve representing the kernel density estimation obtained via ggplot2 in R. Subfigure (b) displays the QQ-plot of $\sqrt{n}(\hat{\alpha} - \alpha_0)/\hat{\sigma}$ along with the quantiles of $\mathcal{N}(0, 1)$, where $\hat{\sigma}$ is the standard deviation of $\sqrt{n}(\hat{\alpha} - \alpha_0)$ over the Monte Carlo iterations. The figures demonstrate the proximity of the distribution of $\sqrt{n}(\hat{\alpha} - \alpha_0)$ to that of $\mathcal{N}(0, \sigma^2)$, validating our theoretical findings.

4.2. Simulation for growing dimension

In this subsection, we present our simulation for a high-dimensional model where we set $n = 5000$, $p = 6000$, and $s = 3$. Our method requires a relatively large n for the asymptotic behavior to manifest due to the slow convergence of the non-parametric function b and its derivative. Thus, we opt for a large n while maintaining a computationally feasible scenario. Our simulation setup is as follows:

1. The true parameter $\beta_0, \gamma_0 \in \mathbb{R}^p$ and with first $s = 3$ co-ordinates being non-zero. The non-zero co-ordinates are generated from $\mathsf{unif}([1, 2])$ and are kept fixed for all Monte Carlo iterations. The value of $\alpha_0 = 2$.
2. The function $b(x)$ is set to be $b(x) = x^3/3$.
3. The co-ordinates of X are generated independently from $\mathsf{unif}([-1, 1])$.
4. We set $Z = X$ and fix $\tau_0 = 0$ to generate Q.



(a) Histogram of $\sqrt{n}(\hat{\alpha} - \alpha_0)$

(b) Q-Q plot

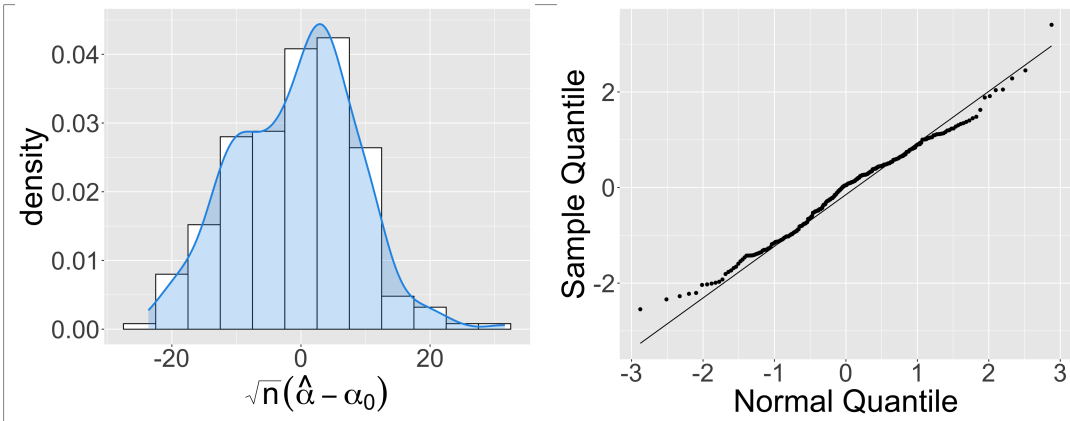
Figure 1: Histogram and Q-Q plot of $\sqrt{n}(\hat{\alpha} - \alpha_0)$ over 500 Monte Carlo iterations in fixed dimension

5. The errors (η, ϵ) are generated independently from $\mathsf{unif}([-1, 1])$.

We generate n data from the aforementioned data generation model and repeat the procedure 500 times (Monte Carlo iterations), keeping all parameters fixed. Similar to Figure 1, Figure 3 presents a comprehensive representation of the resulting estimator. Once again, we have two subfigures: subfigure (a) displays the histogram of $\sqrt{n}(\hat{\alpha} - \alpha_0)$ over 250 Monte Carlo iterations, illustrating a close resemblance to a Gaussian distribution. Subfigure (2) depicts a Q-Q plot comparing $\sqrt{n}(\hat{\alpha} - \alpha_0)/\hat{\sigma}$, where $\hat{\sigma} = \mathrm{sd}(\sqrt{n}(\hat{\alpha} - \alpha_0))$ across the Monte Carlo iterations, with the quantiles of the standard normal random variable. This plot confirms the proximity between the distribution of $\sqrt{n}(\hat{\alpha} - \alpha_0)$ and Gaussian distribution, as supported by our theoretical analysis; however, the proximity is not as good as in the low dimensional case. A careful Berry-Essen-type analysis is required to understand this convergence. The RMSE ($\sqrt{\mathbb{E}[(\hat{\alpha} - \alpha_0)^2]}$) of our experiment is 0.1314559 (where the expectation is approximated by the average of the outputs of 500 Monte Carlo iterations).

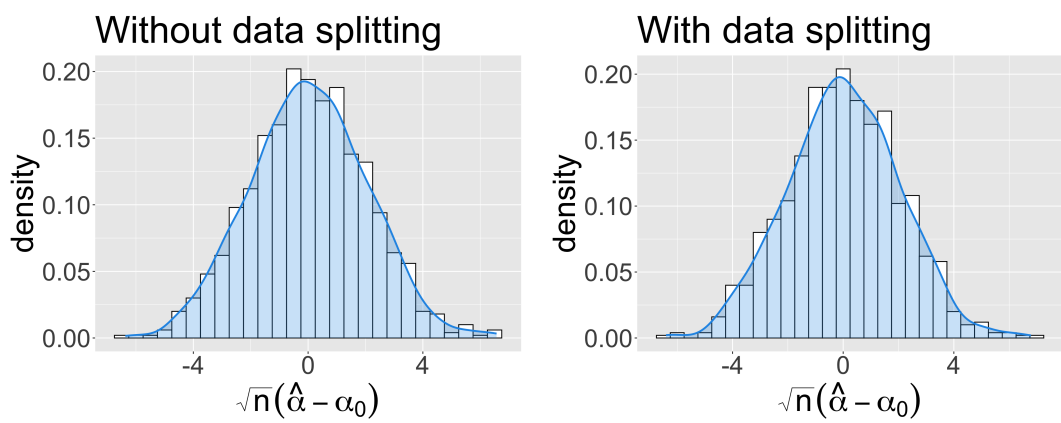
4.3. Simulation in fixed dimension comparing data splitting with no splitting

In this subsection, we present our findings regarding the performance of our algorithm with or without data splitting. Towards that end, we borrow the same setup as for the simulation in fixed dimension (i.e., Section 4.1). However, we run two methods: i) we run the proposed method (i.e., divide our data into three parts) and compare it with ii) the method with no data splitting (i.e., we estimate γ_0, b' and the final estimator from the same set of observations). We have run 1000 Monte Carlo iterations for the purpose of simulation. Here are the histograms (along with kernel density plot) of $\sqrt{n}(\hat{\alpha} - \alpha_0)$ both with and without data splitting: The above figure shows that both methods perform equally well, and we couldn't distinguish between them using simple measures (e.g., histogram, summary statistics, plot). The approximate RMSE (an approximation of $\sqrt{\mathbb{E}[(\hat{\alpha} - \alpha_0)^2]}$ over 1000 Monte Carlo iterations) obtained by data splitting is 0.02853055, whereas the value is 0.02876631 without data splitting. Therefore, we conjecture that our method can be implemented without data splitting; however, data splitting is particularly helpful for establishing theoretical guarantees.



(a) Histogram of $\sqrt{n}(\hat{\alpha} - \alpha_0)$ (b) Q-Q plot

Figure 2: Histogram and Q-Q plot of $\sqrt{n}(\hat{\alpha} - \alpha_0)$ over 250 Monte Carlo iterations in high dimension



(a) Histogram of $\sqrt{n}(\hat{\alpha} - \alpha_0)$ (b) Q-Q plot

Figure 3: Histogram of $\sqrt{n}(\hat{\alpha} - \alpha_0)$ over 1000 Monte Carlo iterations under the setup of Section 4.1 both with and without data splitting.

5. Real data analysis

In this section we illustrate our method by analyzing two real datasets in two subsections.

5.1. Effect of Islamic party on women's education in Turkey

In this subsection, we study the effect of Islamic party rule in Turkey on women's empowerment in terms of their high school education. In the 1994 municipality elections, Islamic parties won several municipal mayor seats in Turkey. We are interested in investigating whether this winning had any effect

on the education of women, i.e., to determine, statistically, whether the concern that Islamic control may be inimical towards gender equality is supported by the data. The dataset we analyzed here was collected by the Turkish Statistical Institute and was first analyzed in Meyersson (2014). Since then it has been used by several authors, appearing for example, as one of the core illustrations in Cattaneo, Idrobo and Titiunik (2019)[3]. The dataset consists of $n = 2629$ rows where the rows represent municipalities, the units of our analysis. The main target/response variable Y is the percentage of women in the 15-20 year age group who were recorded to have completed their high school education in the 2000 census. As mentioned in Meyersson (2014), this group should have been most affected by the decisions made by the winners of the 1994 election. The treatment determining variable Q is the difference in vote share between the largest Islamic party (i.e., the Islamic party which got the maximum votes among all Islamic parties) and the largest secular party (i.e., the non-Islamic party which got the maximum votes among all non-Islamic parties). Hence, the cutoff is 0: i.e., if $Q_i > 0$, then the ith municipality unit elected an Islamic party, and if $Q_i < 0$, a secular party. The available co-variates are described in Table 1 of Mukherjee, Banerjee and Ritov (2023). For X and Z, we use all the co-variates presented in that table except `i89` because almost 1/3'rd of the observations (729 many) are missing for this variable. We estimate α_0 by $\overline{\hat{\alpha}}$ as described in Section 2.1. To test:

$$H_0 : \alpha_0 = 0 \quad vs \quad H_1 : \alpha_0 \neq 0 \,,$$

we construct an Efron-bootstrap-based confidence interval over 500 iterations. We present our findings in Table 1. From the table, it is clear that we do not have enough evidence to reject H_0 at 5% level as the confidence interval contains 0. Hence, we conclude that there is no significant effect of the Islamic ruling party on women's education.

We next compare our result to that of Meyersson (2014) and Cattaneo, Idrobo and Titiunik (2019). Meyersson (2014) implemented a simpler model for RDD:

$$Y_i = \beta_0 + \alpha_0 S_i + f(Q_i) + \epsilon_i \,,$$

with f being a polynomial function, and only those observations were used where $Q_i \in (-h, h)$ for some optimal choice of the bandwidth h (chosen according to the prescription of Imbens and Kalyanaraman (2012)). The authors found that Islamic party rule has a significantly positive effect on women's education at the 1% level test. On the other hand, Cattaneo, Idrobo and Titiunik (2019) implemented the standard RDD model. We replicate their result using the R-package *rdrobust* as advocated in Cattaneo, Idrobo and Titiunik (2019). The function `rdrobust` inside the package *rdrobust* takes input Y, Q, X and splits out three different types of the estimate of α_0 (along with 95% confidence interval), namely `conventional, bias-corrected` and `Robust`. As mentioned in Calonico (2020), `conventional` means standard/traditional RD estimates (i.e., solving RDD via local polynomial regression) with conventional standard errors, `bias-corrected` implies bias-corrected RD estimate with conventional standard errors and `robust` indicates bias-corrected RD estimates with robust standard errors (see Calonico, Cattaneo and Titiunik (2014)). We use the parameters `kernel` = 'triangular', `scaleregul` = 1, `p` = 1, `bwselect` = 'mserd' of rdrobust function to run the analysis. As evident from Table 2, all these estimates reject H_0 at 5% level, stipulating a strictly positive effect of the Islamic party on the education of women, while our method fails to reject the null at the same level.

The key difference between our analysis and the previous method lies in that the previous analysis only considered those municipalities where the Islamic ruling party *barely* won or lost, and based on

[3] We have downloaded the dataset from https://github.com/rdpackages-replication/CIT_2019_CUP/blob/master/CIT_2019_Cambridge_polecon.csv.

Table 1. Summary Statistics of data on Islamic party based on our method

Point Estimate	0.4071513
Bootstrap mean.	0.5760144
Bootstrap s.e.	0.48115
Bootstrap 95% C.I.	$(-0.4234894, 1.41942)$

Table 2. Summary Statistics of data on Islamic party based on Cattaneo, Idrobo and Titiunik (2019)

Name of methods	Coeff	CI Lower	CI Upper
Conventional	3.005951	0.9622239	5.049678
Bias-Corrected	3.204837	1.1611103	5.248564
Robust	3.204837	0.8266720	5.583003

their analysis of those regions, concluded that the Islamic ruling party had a positive effect on women's empowerment; whilst ours is very different in that it uses information from *all* municipalities, i.e., it uses substantially more data. We believe this to be the main reason behind the discrepant outcomes and insist, as before, that in data scenarios of this sort, it makes more sense to extract information from all 'treated' individuals and not merely the ones near the boundary.

5.2. Effect of probation on subsequent GPA

We next analyze an educational dataset, originally collected and analyzed in Lindo, Sanders and Oreopoulos (2010)[4]where we investigate whether putting students on academic probation due to grades below a pre-determined cutoff has any effect on their subsequent GPA. The data are based on students from 3 independent campuses of a large Canadian university – a major campus and other satellite campuses. The acceptance rate in the major campus is around 55% and in the satellite campuses around 77%. The data were collected over 8 cohorts of students till the end of the 2005 academic year. To observe the students for at least two years, only those who entered the university prior to the beginning of the 2004 academic year were considered. After being put on academic probation in their first year, some students left the university. We, therefore, only have access to GPA during the second year for those students who stayed. Thus, our Y variable is the GPA of the first academic term in the second year and the treatment S is whether the student was put on probation. The treatment determining variable Q

[4]We have collected the dataset from https://www.openicpsr.org/openicpsr/project/113751/version/V1/view;jsessionid=A6C09FD5CD7DB8E18EAA77B75BD893B2.

Table 3. Summary Statistics of data on GPA data

Point Estimate	0.2733371
Bootstrap mean.	0.2817404
Bootstrap s.e.	0.016672
Bootstrap 95% C.I.	$(0.248588, 0.3158934)$

is the difference between the first year GPA and the cutoff for academic probation: if $Q < 0$, the student is put on academic probation; otherwise, it is not. The covariates we consider here ($X = Z$) are presented in Table 2 of Mukherjee, Banerjee and Ritov (2023). In Table 3, we summarize our result. It is immediate from the bootstrap confidence interval from Table 3 that for testing $H_0 : \alpha = 0$ vs $H_1 : \alpha \neq 0$, we have enough evidence to Reject H_0 at the 5% level and conclude that the students who are put on academic probation and continue with their education, tend to improve their performance (note that the estimated α_0 is positive) in the subsequent academic year. This makes sense, as the students who did not leave university after being put on academic probation must have a strong incentive to work harder so that they are not expelled from the university. Our findings are in harmony with those obtained in Lindo, Sanders and Oreopoulos (2010), where the author also found the treatment effect to be significant at the 5% level.

Remark 5.1. *Note that in Table 1 and Table 3, we presented bootstrap confidence intervals for the treatment effect instead of asymptotic confidence intervals. This is because consistent estimation of the asymptotic variance of our estimator is not straightforward. Recall that from Theorem 2.8, the asymptotic variance of our estimator is* $e_1^\top \Omega_\tau^{-1} \Omega_\tau^* \Omega_\tau^{-1} e_1$. *As mentioned in Step 1 of the sketch of the proof of Theorem 2.8,* $\mathbf{W}^\top \mathsf{proj}^\perp_{\tilde{\mathbf{N}}_{K,a}} \mathbf{W}/n$ *is a consistent estimator of* Ω_τ *but it is hard to estimate* Ω_τ^* *consistently, which forces us to resort to the bootstrap confidence interval.*

6. Conclusion and possible extensions

In this paper, we proposed a new approach to estimate a non-randomized treatment effect at the $\sqrt{n}$ rate and showed that under homoscedastic normal errors, our method is semiparametrically efficient. We also pointed out in Remark 2.11 that one can use weighted least squares instead of ordinary least squares to take care of heterogenous errors. However, the normality assumption is necessary for semiparametric efficiency as we use least squares to estimate the treatment effect. We now discuss some natural extensions of our models that are worth analyzing as potential future research problems.

6.1. Non-constant treatment effect

Consider the following extension of our model with non-constant treatment effect:

$$Y_i = \alpha(\eta_i)\mathbb{1}_{Q_i>0} + X_i^\top \beta_0 + \nu_i = \alpha(\eta_i)\mathbb{1}_{Q_i>0} + X_i^\top \beta_0 + b(\eta_i) + \epsilon_i \,, \tag{6.1}$$

$$Q_i = Z_i^\top \gamma_0 + \eta_i \,, \tag{6.2}$$

where as before $b(\eta_i) = \mathbb{E}[\nu \mid \eta_i]$. This generalization assumes that the response of a treated candidate with higher abilities is boosted compared to another treated candidate with lower ability. For example, a more capable student, upon entering a prestigious grad school, will most likely get a better mentor, amplifying their academic prowess. The random variable $\alpha(\eta)$ represents the *conditional treatment effect* as:

$$\mathbb{E}[Y \mid X, \eta, Q > 0] - \mathbb{E}[Y \mid X, \eta, Q < 0] = \alpha(\eta)\,,$$

i.e., conditioning on (X, η) (background information and innate ability), $\alpha(\eta)$ quantifies the difference between the responses of treated and untreated samples. It can be shown that the estimator proposed in our manuscript based on the model with constant α_0, estimates, in the newly proposed model, a *weighted average* of $\alpha(\eta)$, i.e. $\mathbb{E}[p(\eta)\alpha(\eta)]/\mathbb{E}[p(\eta)]$ where $p(\cdot)$ is a non-negative integrable function depending on other parameters. This weight function is basically a function of the information for α in each η - stratum. So, our current method is applicable when the parameter of interest is a *weighted average treatment effect*.

However, if the parameter of interest is *unweighted average treatment effect* $\mathbb{E}[\alpha(\eta)]$, then our method can be slightly modified as follows to yield a $\sqrt{n}$-consistent estimator:

1. Split the data into three (almost) equal parts, say $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$.
2. From $\mathcal{D}_1$, impute $\hat{\eta}$ from equation (6.2) by regressing Q on X.
3. Estimate $\alpha(\cdot)$ and $b(\cdot)$ (and their derivatives) from $\mathcal{D}_2$ using a B-spline series expansion.
4. Note that equation (6.1) can be expanded as:

$$\begin{aligned}
Y_i &= \alpha(\eta_i)\mathbb{1}_{Q_i>0} + X_i^\top \beta_0 + b(\eta_i) + \epsilon_i \\
&= \alpha(\hat{\eta}_i)\mathbb{1}_{Q_i>0} + b(\hat{\eta}_i) + X_i^\top \beta_0 + (\hat{\gamma} - \gamma_0)^\top Z_i \left(\alpha'(\hat{\eta}_i)\mathbb{1}_{Q_i>0} + b'(\hat{\eta}_i)\right) + \epsilon_i + R_i \\
&\approx \alpha(\hat{\eta}_i)\mathbb{1}_{Q_i>0} + b(\hat{\eta}_i) + X_i^\top \beta_0 + (\hat{\gamma} - \gamma_0)^\top Z_i \left(\hat{\alpha}'(\hat{\eta}_i)\mathbb{1}_{Q_i>0} + \hat{b}'(\hat{\eta}_i)\right) + \epsilon_i + R_i\,,
\end{aligned}$$

where in the last line we replaced α' and b' by their estimates obtained from $\mathcal{D}_2$ in the previous step. Finally, we can re-estimate α from the above equation via a non-parametric method (i.e., expanding via a B-spline basis and regressing Y on an appropriate set of covariates). Our final estimator becomes:

$$\hat{\mathbb{E}}[\alpha(\eta)] = \frac{1}{n}\sum_{i=1}^{n} \hat{\alpha}(\hat{\eta}_i)\,.$$

We conjecture that this estimator has $\sqrt{n}$ rate of convergence and is asymptotically normal.

One may also analyze this heterogeneous treatment effect model when the links are non-linear, i.e., say $Q = g(Z) + \eta$ for some non-linear function g. However, the trade-off is that we require more smoothness on $b(\eta)$ in (6.1). The key reason is that, in the presence of a linear link, i.e. $Q = Z^\top \gamma_0 + \eta$, the residuals can be approximated with $n^{-1/2}$ error rate, whereas for the non-linear link, $\hat{\eta} = Q - \hat{g}(Z)$ can only approximate η at a non-parametric rate (which is always larger than $n^{-1/2}$), and the rate depends on the smoothness of g. This, in turn, affects the estimation of $b(\cdot)$; previously, as the true η was approximated with the error of the order $n^{-1/2}$, we only needed to estimate b and $b'()$, as the contribution of higher order derivatives is smaller than $n^{-1/2}$ and asymptotically negligible. However, when η is approximated at a non-parametric rate slower than $n^{-1/2}$, we might need to estimate two or more derivates of b (depending on the smoothness of g as this determines the approximation error of η).

6.2. Bootstrap consistency

As noted in Remark 5.1, we use a bootstrap confidence interval instead of the asymptotic one owing to the intricate form of the asymptotic variance of our estimator, which makes it hard to estimate from the data. Therefore, an immediate question of interest is to investigate whether the bootstrap is consistent under our model assumptions. Although empirical evidence suggests that this is the case, a rigorous theoretical undertaking is essential to establish the claim.

Supplementary Material

Supplement to "Estimation of a score-explained non-randomized treatment effect in fixed and high dimensions"
The supplementary material contains the proofs of the Propositions and Theorems, some discussion on the semiparametric efficiency, and some relevant descriptions of the real data analysis.

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